

碳和硅相关化学习题册

Carbon and Silicon Related Chemistry Exercise Book

一、选择题

I. Multiple-choice Questions

1. 下列关于 CO 的说法中, 不正确的是()

Which of the following statements about CO is incorrect? ()

A. CO 有剧毒/ CO is highly toxic.

B. CO 易溶于水/ CO is highly soluble in water.

C. CO 在空气中燃烧时的火焰为淡蓝色/ The flame of CO burning in air is pale blue.

D. CO 可在高温下还原氧化铁/ CO can reduce iron oxide at high temperatures.

2. 鉴别 CO₂, 常用的方法是()

The common method to identify CO₂ is ()

A. 通入水中, 溶液变为酸性/ Pass it into water and the solution becomes acidic.

B. 通入澄清石灰水中, 石灰水变混浊/ Pass it into clear lime water and the lime water becomes turbid.

C. 与镁反应生成碳和氧化镁/ React with magnesium to form carbon and magnesium oxide.

D. 与碳在高温下反应生成 CO/ React with carbon at high temperatures to form CO.

3. 向澄清的石灰水里通入二氧化碳, 石灰水变混浊, 这是因为生成了白色的()

When carbon dioxide is passed into clear lime water, the lime water becomes turbid because a white () is formed.

A. BaCO₃

B. CaCO_3

C. $\text{Ca}(\text{OH})_2$

D. CaO

4. 不能将黑色氧化铜还原为红色金属铜的是()

What cannot reduce black copper oxide to red metallic copper is ()

A. CO

B. CO_2

C. H_2

D. C

5. 实验室里制取二氧化碳的反应是()

The reaction for producing carbon dioxide in the laboratory is ()

A. $\text{C} + \text{O}_2 = \text{CO}_2$

B. $2\text{CO} + \text{O}_2 = 2\text{CO}_2$

C. $\text{CaCO}_3 + 2\text{HCl} = \text{CaCl}_2 + \text{H}_2\text{O} + \text{CO}_2 \uparrow$

D. $\text{CO} + \text{CuO} = \text{Cu} + \text{CO}_2$

6. 下列关于硅及其化合物的说法中, 正确的是()

Which of the following statements about silicon and its compounds is correct? ()

A. 硅在自然界中只以化合态的形式存在/ Silicon exists only in the form of compounds in nature.

B. 硅在地壳中的含量在所有元素中排第一位/ Silicon ranks first among all elements in terms of its content in the earth's crust.

C. 硅酸的酸性比碳酸强/ The acidity of silicic acid is stronger than that of carbonic acid.

D. SiO_2 不与任何酸反应/ SiO_2 does not react with any acid.

7. 要除去一氧化碳中混有的少量二氧化碳，可采用的方法是()

To remove a small amount of carbon dioxide mixed in carbon monoxide, which of the following methods can be adopted? ()

- A. 把混合气体点燃/ Ignite the mixed gas.
- B. 使混合气体通过灼热的氧化铜粉末/ Pass the mixed gas through the hot copper oxide powder.
- C. 把混合气体由一个容器倾倒到另一个容器中/ Pour the mixed gas from one container to another.
- D. 使混合气体通过过量的澄清石灰水/ Pass the mixed gas through excessive clarified lime water.

8. 一氧化碳具有毒性是因为()

Carbon monoxide is toxic because ()

- A. 它具有还原性/ It has reducibility.
- B. 它具有可燃性/ It has combustibility.
- C. 它能与血液里的氧气结合生成二氧化碳/ It can combine with oxygen in the blood to form carbon dioxide.
- D. 它能与血液里的血红蛋白牢固结合/ It can firmly combine with hemoglobin in the blood.

9. 下列关于 CO_2 的说法中，不正确的是()

Which of the following statements about CO_2 is incorrect? ()

- A. CO_2 可溶于水，形成酸性溶液/ CO_2 is soluble in water and forms an acidic solution.
- B. CO_2 可在高温下还原氧化铜/ CO_2 can reduce copper oxide at high temperatures
- C. CO_2 可与氢氧化钙发生反应/ CO_2 can react with calcium hydroxide

D. CO₂ 在空气中既不燃烧也不支持燃烧，所以一般可用作灭火剂/ CO₂ neither burns nor supports combustion in air, so it is generally used as a fire extinguishing agent

二、填空题

II. Fill in the blanks

1. 固体二氧化碳称为_____。

Solid carbon dioxide is called_____. (The specific answer is not provided in the original text, so it cannot be accurately filled in here)

2. 碳在氧气中充分燃烧，生成_____；碳在氧气中不充分燃烧，生成_____。

When carbon burns sufficiently in oxygen, it produces_____; when carbon burns insufficiently in oxygen, it produces_____. (The specific answers are not provided in the original text, so they cannot be accurately filled in here)

3. 由于二氧化碳_____，因此不能用排水法收集；利用二氧化碳_____，我们可以采用向上排空气法来收集二氧化碳。

Since carbon dioxide_____, it cannot be collected by the drainage method; using the fact that carbon dioxide_____, we can use the upward air displacement method to collect carbon dioxide. (The specific properties are not provided in the original text, so they cannot be accurately filled in here)

4. 一氧化碳在高温下还原铁矿石里的氧化铁，生成_____和_____。

The iron ore is reduced by carbon monoxide at high temperatures, producing_____ and_____. (The specific products are not provided in the original text, so they cannot be accurately filled in here)